

Exercises: Deep learning with Pytorch

Exercise 1: Function Approximation of $\sin(x)$

Goal: Train a neural network to approximate the function $f(x) = \sin(x)$.

- a) Generate training data by sampling $x \in [0, 2\pi]$ and computing $y = \sin(x) + \text{noise}$.
- b) Build a feedforward neural network that maps x to y .
- c) Train the network using MSE loss and the Adam optimizer.
- d) Plot the learned function alongside the true $\sin(x)$ curve.
- e) Plot the losses over the training epochs

Exercise 2: Time-Dependent Function Approximation

Goal: Adjust the algorithm from the first exercise to learn a time-dependent function $f(t, x) = \sin(x) \cdot \cos(t)$ using a neural network. You don't have to plot the learned function.

Exercise 3: Optimization over Functions

Goal: Use a neural network to learn a function $x = u(y)$ that minimizes a cost functional $g(x, y) = (x - \sin(y))^2$, where $y \in [-\pi, \pi]$.